SFWRENG / MECHTRON 3K04 Software Development

Today: Midterm Review Session

McMaster University | Engineering Computing & Software

October 20th, 2026

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Midterm: Friday, October 23rd 11:30 AM – 12:20 PM

- **BE ON TIME! (Arrive early)**
- Two rooms
  - **T13 123 -> Last Names A-L**
  - **T13 127 -> Last Names M-Z**
- 45 minutes
- **One cheat sheet is allowed. Letter sized. Double-sided.**
- Types of questions:
  - Multiple-choice for *theory*
  - Short answers for *practice*
    - *You practice by working on your project!*

Requirement Analysis

A university is developing a **Smart Study Room Booking System (SRBS).** Students can reserve study rooms, ch eck availability, and cancel reservations. University staff can view usage reports.

The development team has proposed these requirements:

a) The system shall allow students to reserve rooms quickly and easily.

b) The system shall send an email notification and update the room's availability when a reservation is cancelled.

c) The system shall use the university's servers to store reservation information.

For each requirement:

- Identify one potential requirements-quality problem.
- Rewrite the requirement to address the problem.
- Classify your rewritten requirement as functional (FR) or non-functional (NFR).

User Story and Acceptance Criteria

The SRBS plans to introduce an automatic room reservation cancellation feature.

If a student has not checked into their reserved study room within 15 minutes of the reservation's start time, the sy stem must cancel the reservation and make the room available to other students.

Students who check in before the deadline must retain their reservations.

a) Write a user story for this feature from the perspective of a student or library staff member.

b) Write two acceptance criteria using Gherkin syntax (Given–When–Then). Your criteria should test two meaningfully different situations, including a boundary or exceptional case.

Use Case Modelling

An online banking application supports the following use cases:

- When a customer transfers money, the system must verify that sufficient funds are available before completing the transfer.
- Customers may request a transaction receipt after successfully completing a transfer.

Identify the appropriate <<include>> or <<extend>> relationship between:

- TransferMoney and VerifyAvailableFunds.
- RequestReceipt and TransferMoney.

Briefly justify both choices.

State Behaviour

A parking garage uses an automated barrier system.

The system has four states:

- **Closed**: The barrier is lowered and no vehicle is passing.
- **Opening**: The motor is raising the barrier.
- **Open**: The barrier is fully raised.
- **Fault**: A malfunction has been detected.

The controller receives the following events:

Event	Description
`validTicket`	A driver presents a valid parking ticket
`fullyRaised`	The barrier reaches the upper limit switch
`vehiclePassed`	A vehicle has cleared the barrier
`motorFault`	The motor reports a malfunction
`reset`	An authorized technician requests a reset

Assume that a successful `reset` returns the controller to **Closed**, and `motorFault` can occur in any non-Fault state.

a) Construct a state transition table showing the next state for each applicable event. Assume that `vehiclePassed` in the **Open** state returns the barrier to **Closed**, with closing operations abstracted away.

b) While the barrier is already **Open**, another `validTicket` event is received. Discuss whether this event should be treated as invalid or as a fault. State a reasonable controller response and explain why that response should be specified in the requirements.

Modularity

A team is developing a university course registration system. Its current architecture contains the following modules:

- `StudentInterface`
- `RegistrationProcessor`
- `DatabaseManager`
- `NotificationService`

The team has made the following design decisions:

- `RegistrationProcessor` calls `DatabaseManager` to retrieve course information.
- `DatabaseManager` calls `RegistrationProcessor` to determine whether a student is eligible to register.
- `NotificationService` directly accesses fields inside `DatabaseManager` to obtain student email addresses.

The university expects frequent changes to registration rules, database technology, and notification methods.

a) Identify two design problems involving module dependencies or information hiding. Explain the consequences of each.

b) Propose an improved division of responsibilities that would make the system easier to modify. Explain how your proposal isolates at least two of the anticipated changes.

Preconditions, Postconditions and Invariants

A university laboratory uses an automated equipment cabinet to store expensive electronic instruments. Students must authenticate before removing equipment. The cabinet controller maintains the following variables:

Variable	Description
`state`	`Locked`, `Unlocked`, or `Alarm`
`authenticated`	True if the student has successfully authenticated
`doorClosed`	True if the cabinet door is closed
`alarmActive`	True if the alarm is active

The controller provides an operation `unlockCabinet()` with the following required behaviour:

- The operation may only be requested when the cabinet is in the `Locked` state.
- If the student is authenticated and the door is closed, the cabinet transitions to `Unlocked`.
- If the student is not authenticated, the cabinet remains `Locked` and the request is rejected.
- If the student is authenticated but the door is open, the cabinet transitions to `Alarm` and activates the alarm.

a) Write the precondition for `unlockCabinet()` using the notation introduced in class.

b) Write the postcondition for `unlockCabinet()`, covering all possible outcomes under the precondition. Use logical conditions and implications (⇒) where appropriate.

c) Write one state invariant that prevents unauthorized access to the cabinet.

d) A student proposes the following invariant: `state = Locked ⇒ doorClosed` Is this a valid invariant for the specified system? Justify your answer using the possible system states and transitions.

4-Variable Model

A company is developing an automated water tank controller. The controller must maintain a minimum water level in a storage tank.

The system requirement is: *When the actual water level is below 25 cm, the inlet valve shall be open. Otherwise, the inlet valve shall be closed.*

The system has the following hardware:

- An ultrasonic sensor measures the distance d in cms, between the sensor and the water surface.
- The sensor is mounted 120cm above the bottom of the tank
- The software receives an integer input according to: `i=round(4d).`

The inlet valve is controlled by a digital output: 1 means open and 0 means closed.

a) Identify the M, C, I, and O variables for this system. Clearly distinguish physical quantities from software-visible values.

b) Explain the responsibilities of SOF_in, SOF_req, and SOF_out in a modified four-variable model. Derive an expression that converts the sensor input i into the monitored water level in centimetres.

Questions?